Effects of Innervation on Oligodendrocyte and Microglia Cell Dynamics in the Zebrafish Visual System

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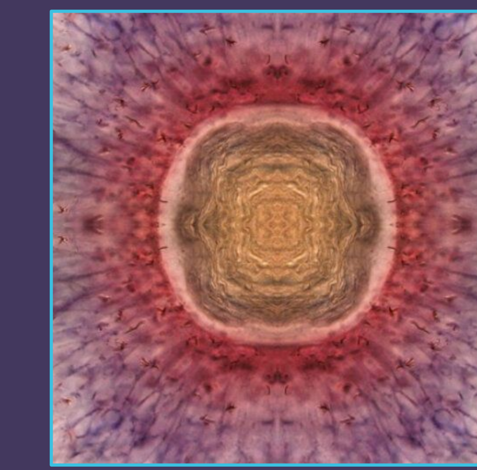
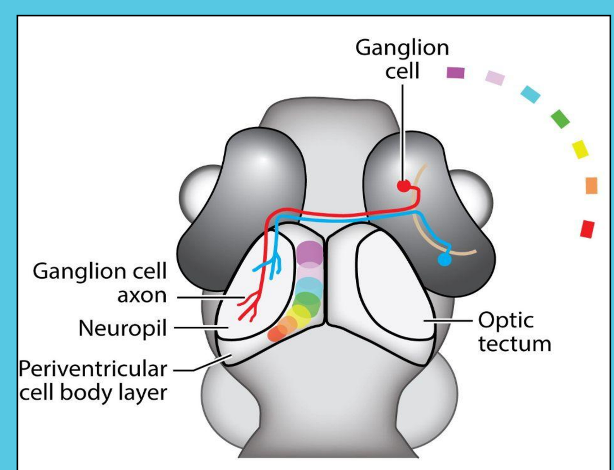


Background



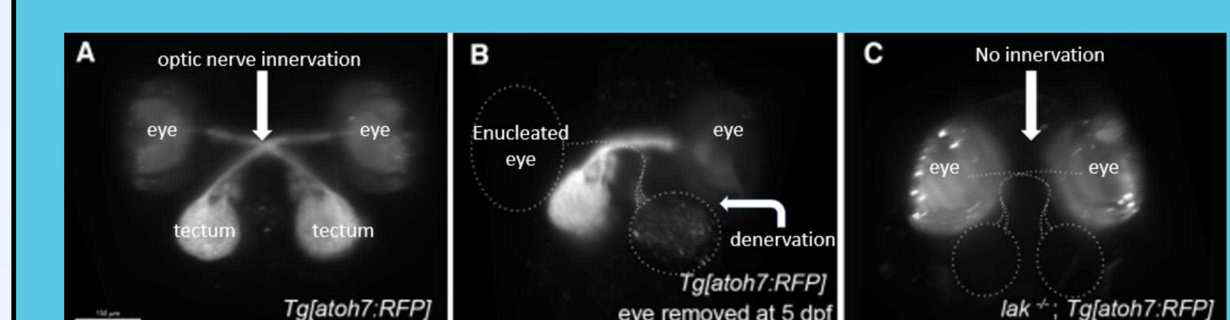
Adapted from Bollman et al., 2019.

The optic nerve innervates the optic tectum (OT) forming the retinotopic map, and is essential for OT development and survival. Denervation leads to reduced growth and more cell death.

Microglia patrol the OT and clear degenerating retinal ganglion cell (RGC) axons after denervation

Oligodendrocyte progenitor cells (OPCs) facilitate proper mapping refinement. They will eventually mature and become mylenating.

Studying OPCs and microglia dynamics helps us learn about OT development, plasticity, and injury response.



WT | Surgical | Genetic

Driving Question

How do microglia and OPC dynamics differ in innervated and denervated OT?

Annexin A5 (apoptosis Mpeg (microglia)

Paired fish with **mpeg:mCherry** (microglia) marker

↓

Injected 1-cell embryos with apoptosis marker (**Annexin A5:GFP**)

↓

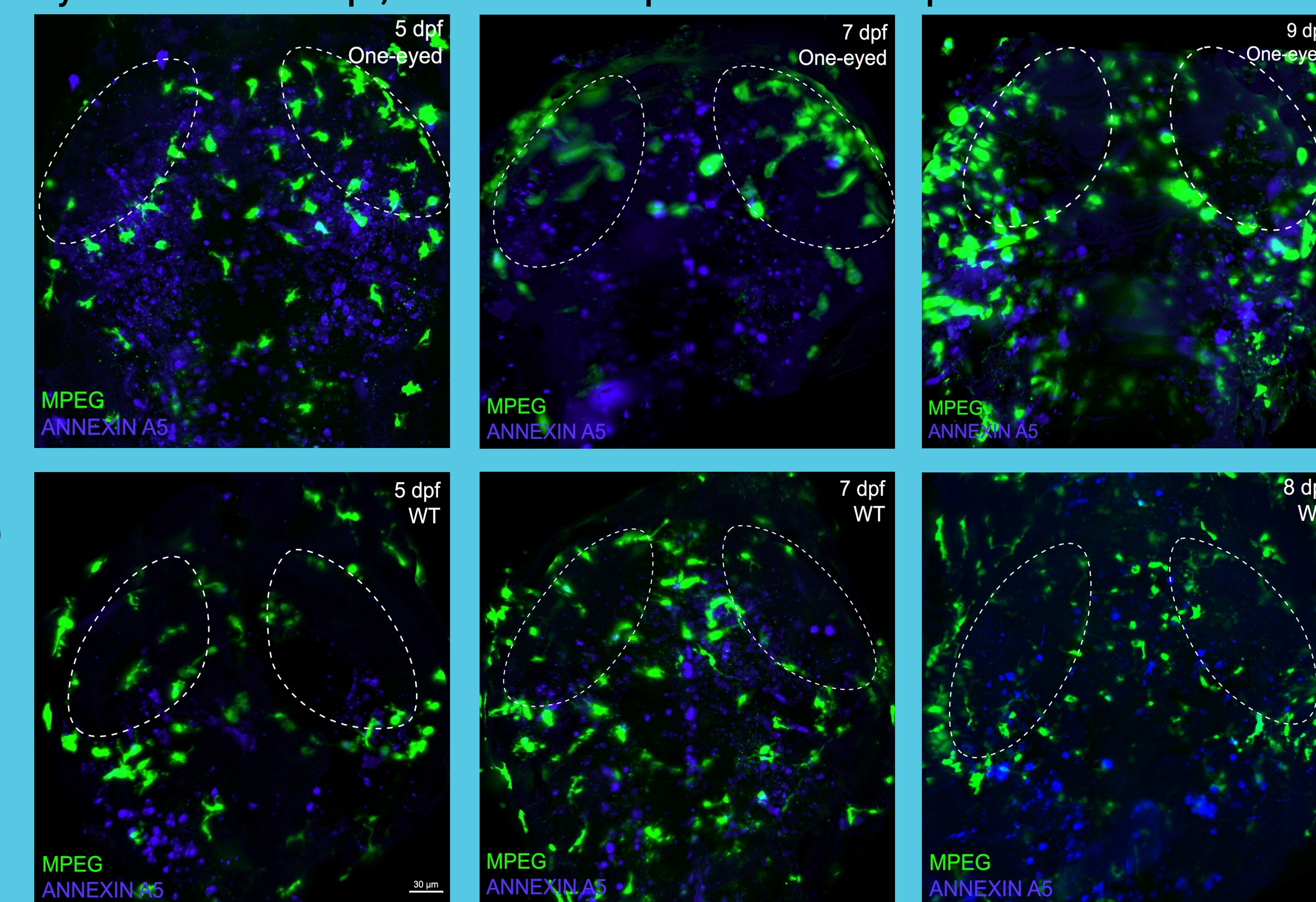
Surgically removed one eye at 5 dpf

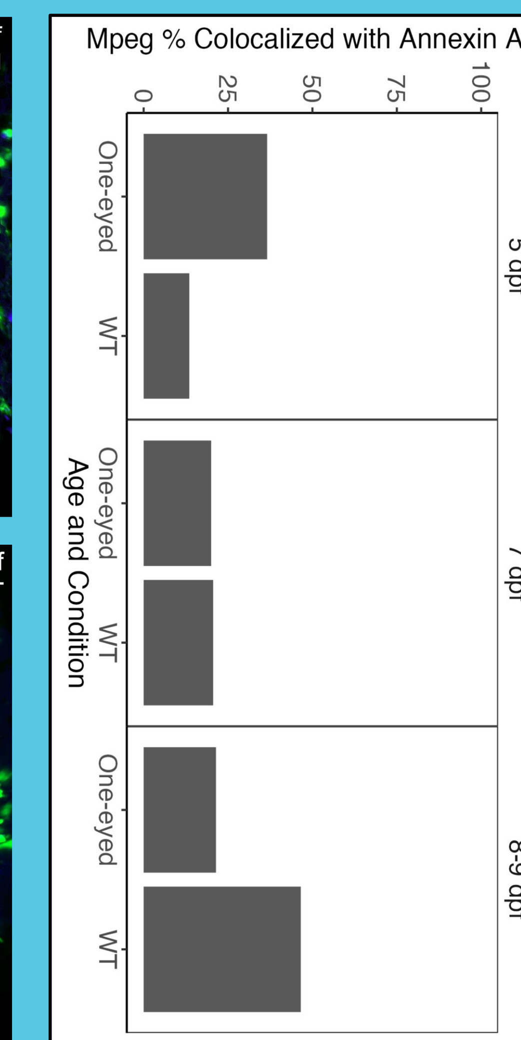
↓

Live time-lapsed between 5 - 9 dpf

- Annexin+ cells can be seen blebbing and fragmenting
- Microglia engulf and clear clusters of apoptotic cells
- Few microglia are colocalized with Annexin A5 across groups
- Microglia morphology seems to differ between groups and ages
- Microglia are preferentially localized in the OT neuropil areas of one-eyed larvae

Microglia fluorescently labeled with mpeg:mCherry (green) and apoptotic cells with Annexin A5:GFP (blue). Live cell images/timelapses were acquired with light sheet microscopy and quantified in Imaris 10.1. Left eye removed at 5 dpf; Dotted ovals represent OT neuropil.





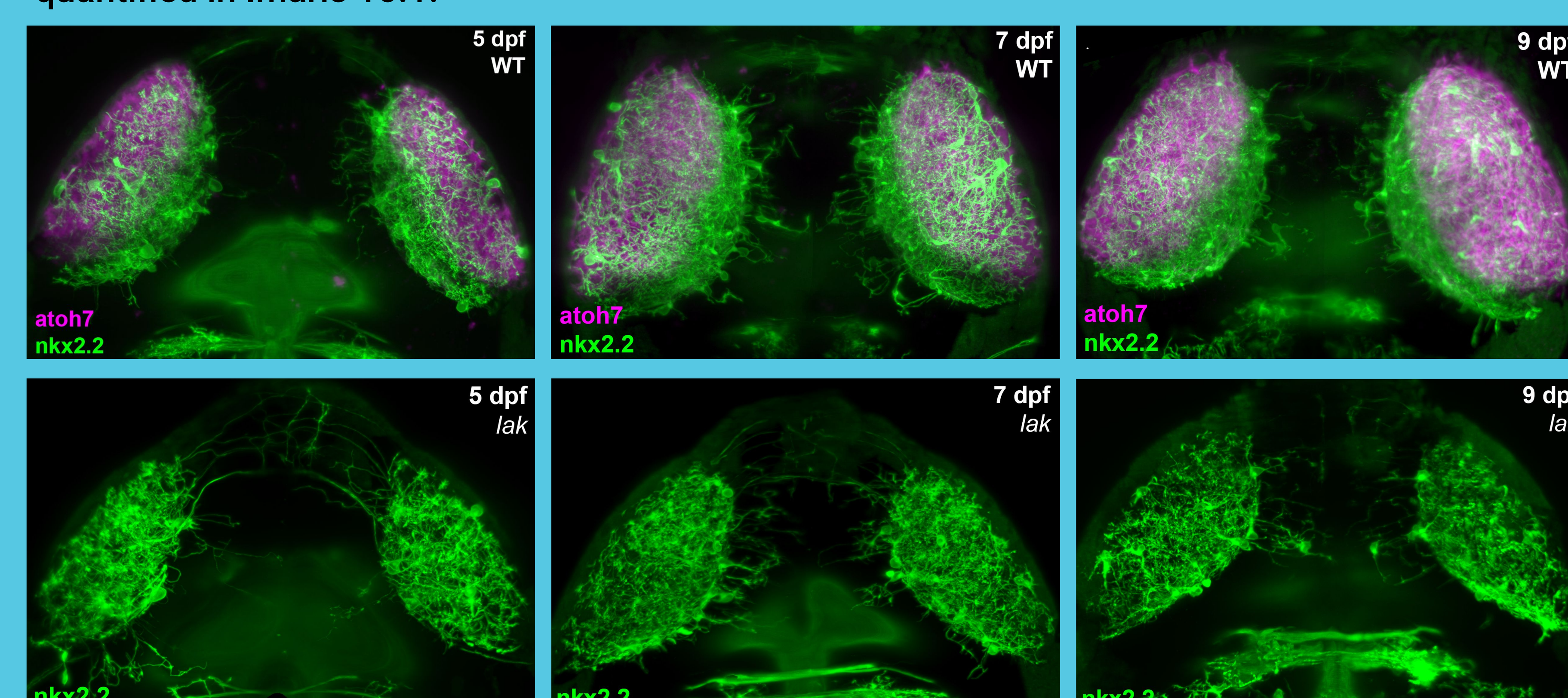
OPCs

Paired *lak +/-* fish with markers for **nkx2.2** and **atoh7** (oligodendrocytes and RGCs)

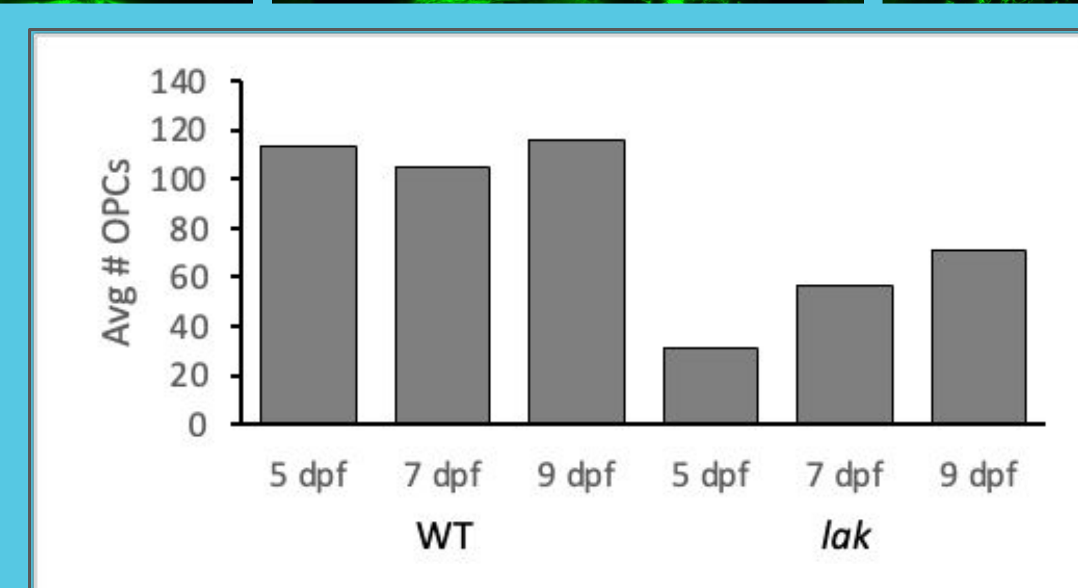
↓

Live time-lapsed between 3 - 9 dpf

Oligodendrocytes fluorescently labeled with nkx2.2:GFP (green) and RGCs with atoh7:RFP (magenta). Live cell images/timelapses were acquired with light sheet microscopy and quantified in Imaris 10.1.



- *lak* mutants fail to develop RGCs and have no optic nerve
- *lak* mutants have different OPC distributions and fewer overall
- *lak* OPCs branch more and project inter- tectally



Average number of OPC cell bodies in tectal lobes across three time points. 1-2 larvae per condition

Conclusions

- Optic nerve denervation promotes microglia phagocytosis
- Microglia display dynamic interactions with apoptotic cells and debris, including phagocytosis
- Denervation leads to microglia injury state
- Un-innervated OTs contain fewer OPCs which are highly arborized
- OPC distribution patterns seem to depend on innervation

Future Directions: More and detailed analyses of timelapses, microglia in *lak*, neuron calcium activity, expanded time frame

Acknowledgements

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